

IDEX Algorithms Template

Draft 8

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Chapter 0

SDC Prompts

All of the information below was taken from the website:

<https://lasp.colorado.edu/galaxy/display/IMAP/IMAP+SDC+Software+Developer+Guide>.

In addition, **from the IMAP SOC CDRL List Applicable Algorithms** is defined in the following sequence:

Level 0 – Level 1A
Level 1A – Level 1B
Level 1B – Level 2
Data culling

0.1 List of Acronyms

- AID: Application Identifier
- APID: Application Process Identifier
- CDRL: Contract Data Requirements List
- CCSDS: Consultative Committee for Space Data Systems
- CoDICE: Compact Dual Ion Composition Experiment
- DAB: Decontamination Actuator Board
- FPGA: Field Programmable Gate Array
- FSW: Flight Software
- HG: High Gain
- IDP: Interplanetary Dust Particle
- IDEX: Interstellar Dust Experiment
- IMAP: Interstellar Mapping and Acceleration Probe
- ISD: Interstellar Dust

- LG: Low Gain
- MG: Mid Gain
- MSPS: Mega samples per second
- PUI: Pickup ions
- SDC: Science Data Center
- SDMP: Science Data Management Plan
- SOC: Science Operations Center
- SWAPI: Solar Wind and Pickup Ions
- TOF: Time of Flight

0.2 Definitions

Algorithm: In this text, algorithm refers to a finite sequence of well-defined, computer-implementable instructions needed to generate one or more data products using lower-level data, calibration, and ancillary information.

Code: Computer-executable instructions in a common programming language may be compiled (if needed) and executed to perform designated computations.

Pseudocode: Simplified code used in preliminary code design and layout.

0.3 Suggested Algorithm Document Content

1. Definitions of Terms
2. Description of Overall Approach
3. Background on Heritage Instruments (if applicable)
4. Background on heritage processing implementations (if applicable)
5. Instrument measurement approach and relationship between data collection, telemetry packets, and data products
6. Overall processing flow, e.g. step-by-step process description/diagrams
7. Identification of and detail pertaining to dependencies on S/C or other instrument's data
8. Calibration Data and associated application, including recommended calibration file formats
9. Mathematical description of algorithm steps
10. Equations, constants, and coefficients for science calibrations
11. Pseudocode
12. Examples of in-flight calibration update incorporation

13. Telemetry packet decomposition information (possibly part of SE-005)
14. Spice kernels to map instrument into spacecraft coordinate frame
15. Equations and coefficients for Temperature Conversions (possibly part of SE-005)
16. Equations and coefficients for engineering conversions (possibly part of SE-005)
17. Observation time basis
18. Expected maintenance tasks (e.g. updating calibrations)
19. Approach to Rejection, Culling, or Flagging of Suspect/Bad Data
20. Expected Uncertainties / Uncertainty Analysis
21. Data Validation Steps
22. Recommended Testing Approaches
23. Special considerations and unusual circumstances

Chapter 1

Introduction

1.1 Document Purpose

This document provides a description of the algorithms used to produce science data products from the IMAP Interstellar Dust Experiment (IDEX) instrument measurements. The data products are documented in the IMAP Science Data Management Plan (SDMP). This document includes a description of the data collection and data products, as well as details of the processing flow, algorithms, and data processing tasks.

1.2 Scope

The intended purpose of this information is to enable members of the IMAP Science Data Center (SDC) to implement, test, and validate executable software that will routinely produce designated data products during the science mission. For IMAP, Science Algorithm Documentation consists of the combination of documentation, instructions, specifications, and any available reference/prototype computer code needed to produce a science data product.

1.3 Document Organization

The first two chapters of the document provide a brief overview of what IDEX is and how it takes measurements. Chapter 2 Introduces the IDEX instrument characteristics and expected performance. Chapter 3 describes the details of the modes of operation: Booting Mode, Idle Mode, Science Mode, Transmit Mode, Safe Mode, Survival Mode and a Decontamination Mode. Chapter 4 describes the necessary steps to convert data between each of the levels along the pipeline. Detailed information on packets and software is made available in the Appendix.

Chapter 2

IDEX Instrument Description

2.1 Overview

The Interstellar Dust Experiment (IDEX) is a high-resolution impact ionization time-of-flight (TOF) mass spectrometer that provides the elemental composition and mass distributions of ISD and IDP particles. IDEX links the interstellar gas phase composition, as obtained with IMAP-Lo and through PUIs with CoDICE and SWAPI, with the makeup of ISD and IDP particles.

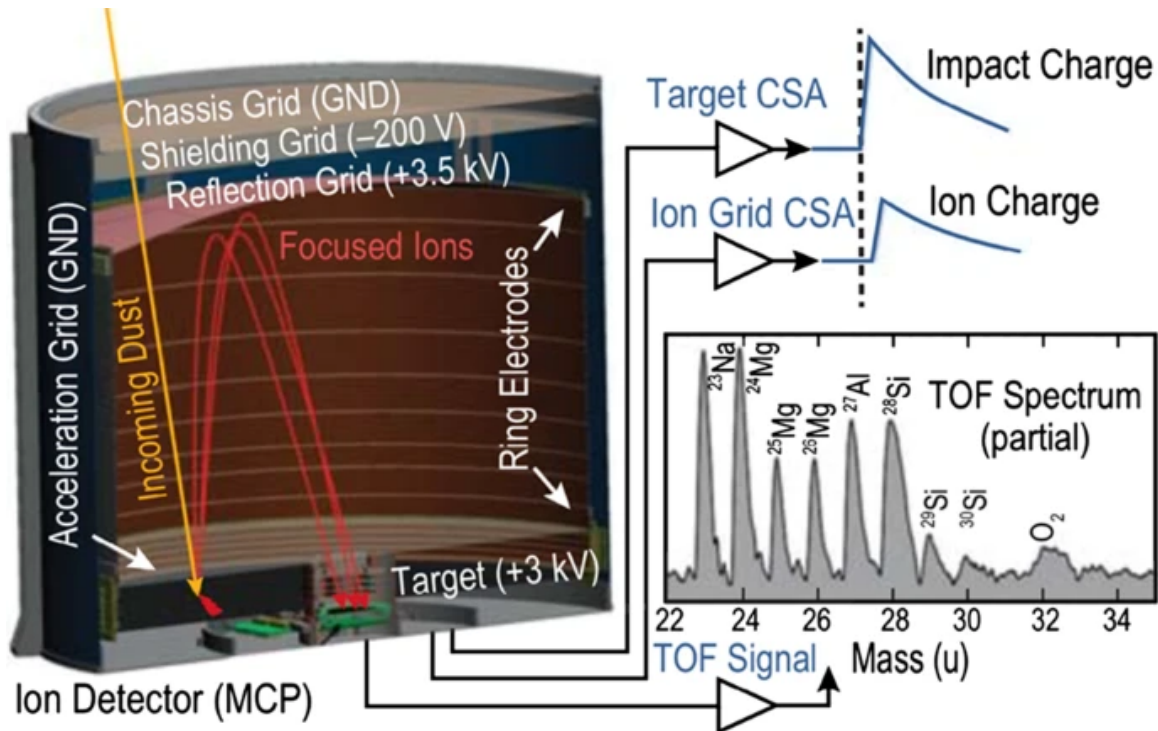


Figure 2.1: An outline of the IDEX instrument and principle of operation. The three types of signals produced (ion grid, target and time-of-flight) are labeled and depicted by their corresponding hardware.

IDEX is based on the heritage of past flight and laboratory instruments. Its operation principle is based on the impact ionization process.

IDEX has two components, an electronics enclosure and a sensor head with a large effective target area (600 cm^2). The IDEX mechanical design consists of a simple Al shell structure that provides support for the biased electrodes of the TOF ion mass analyzer and the grids over its aperture. The centrally-located ion detector and the front-end charge sensitive amplifier are integrated into the bottom of the instrument.

Individual dust particles entering the instrument pass through a set of grid electrodes and impact the target (Figure 2.1). The target is biased at +3 kV to provide positive ion acceleration and the reflectron-type ion optics are used to generate TOF compositional mass spectra. The impact-generated negative charges are collected on the target. An electrostatic field, shaped by a set of biased rings and a curved grid electrode, provides spatial and temporal focusing of the accelerated ions onto the central detector.

Target cleanliness is of high importance to be maintained throughout the fabrication, integration and operation of IDEX. The primary concern is the condensation of volatiles on the target. To mitigate this risk, IDEX has a one-time deployable door and its in-flight operation allows for the regular (once per months) use of heaters to raise the target temperature to 120 C over 8 hours in order to release any possible condensable contaminants.

2.2 Sensor Head

An overview of the sensor head is shown in Figure 2.2.

IDEX Sensor Head Overview

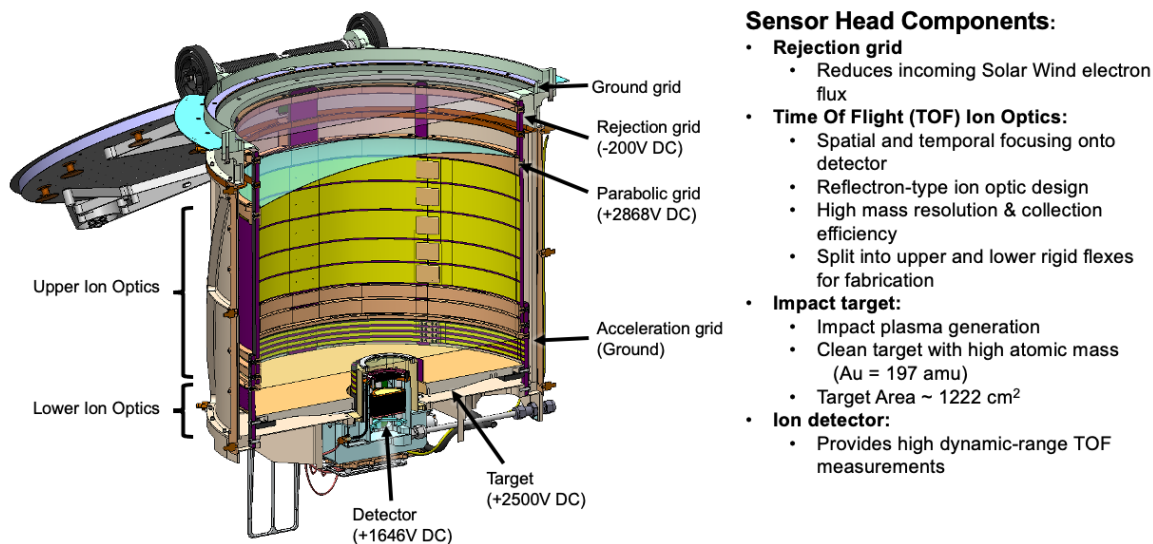


Figure 2.2: An outline of the IDEX instrument sensor head.

2.3 Electronics

The IDEX electronics leverage SUDA flight designs. The Decon/Actuator Board (DAB) is a new design that was added to control the door and decontamination heater. The amplifier boards (CSA, Detector Buffer) were re-designed from the SUDA heritage instrument to meet the IDEX requirements for charge detection ranges. An overview of the IDEX electronics is depicted in Figure 2.3.

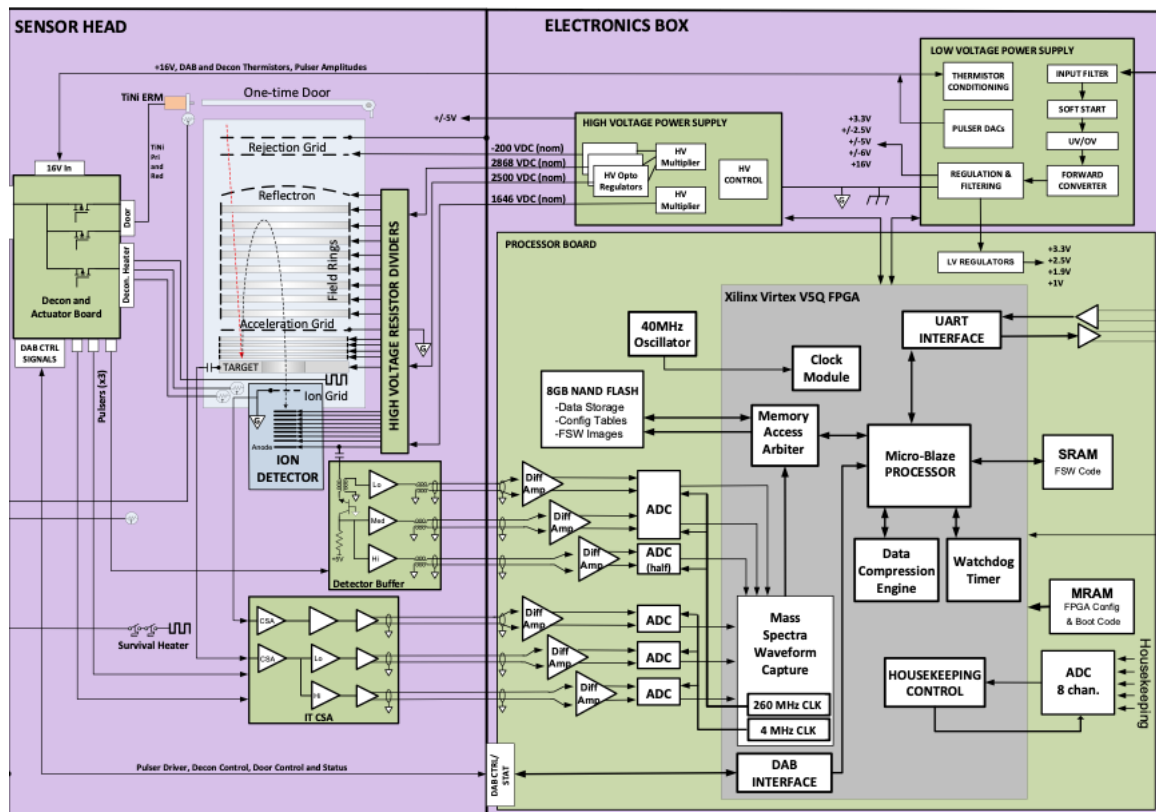


Figure 2.3: An outline of the IDEX instrument electronics system.

The 3 Time-of-Flight (TOF) gain stages, 2 Target CSA and 1 Ion grid CSA waveforms are the signals which the science data products are derived from. These constitute the packetized data accumulated in an 8 GB NAND flash and form the IDEX level 0 data product.

Chapter 3

Nominal Operation

3.1 Operating Modes

IDEX has seven operating modes: off; boot; and five nominal operations. A flow diagram showing how the instrument transitions through the various modes and a brief description of each is depicted in Figure 3.1.

Instrument Modes

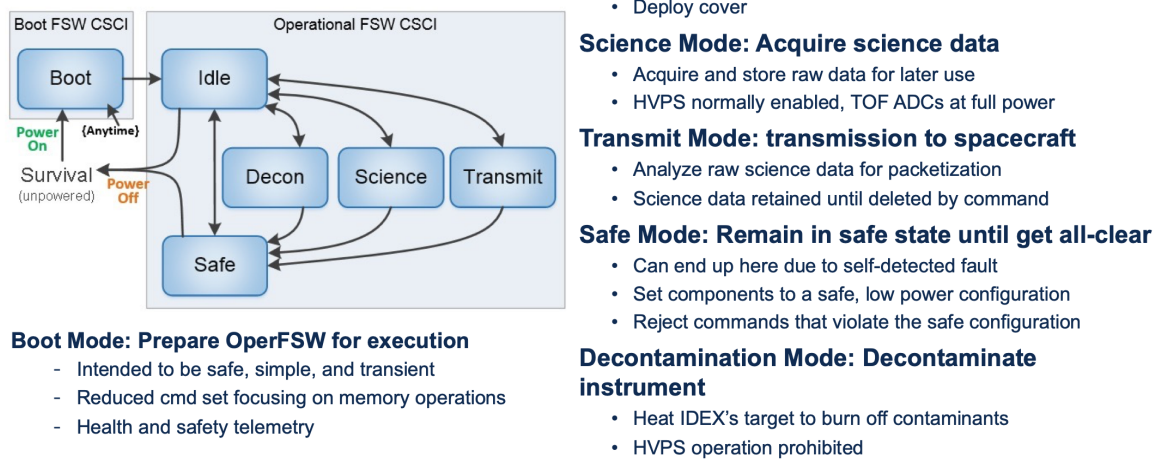


Figure 3.1: An outline of the IDEX instrument operation modes.

The **boot** mode is enabled when the instrument is powered on or any other time the operator indicates. In this mode, while FSW is performing a set of memory operations to prepare for execution, a health and safety telemetry packet is transmitted.

Following boot, the instrument enters its **idle** mode, and the instrument awaits commands. From here, the instrument may enter one of four possible operational modes, of which only two are relevant to the science data processing pipeline.

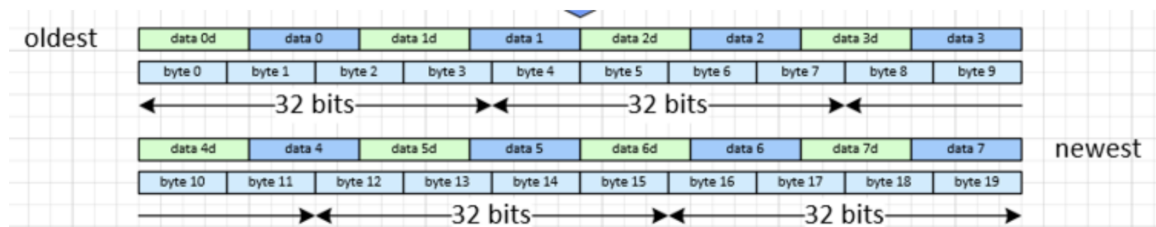
3.1.1 Science Mode

When the instrument enters the **science** mode, a memory buffer is continuously filled containing each of the 6 instrument signals. Since IDEX is an event-driven instrument, data is only packetized and transmitted when we have a dust impact, which we will refer to as an **event**. The instrument triggers based off of an event, sending the data stored in the memory buffer to the NAND flash.

The amount of data stored in a memory buffer depends on the sampling rate of the corresponding **analog-digital converter (ADC)**. The three time-of-flight gain stages are sampled at 260 MHz (high sampling rate). The two target gain stages and the ion grid, the ADCs are sampled at 4 MHz. Figure 3.2 illustrates how data is packed for these two types of ADCs, as well as how much data is transmitted per event.

High Sampling Rate ADCs (TOF Low, Medium and High gain):

- 10-bits per sample
- 260 MHz sampling rate
- 8,192 samples = 32 μ s per event



Low Sampling Rate ADCs (Ion Grid, Target Low Gain, Target High Gain):

- 12-bits per sample
- 4 MHz sampling rate
- 512 samples = 128 μ s per event

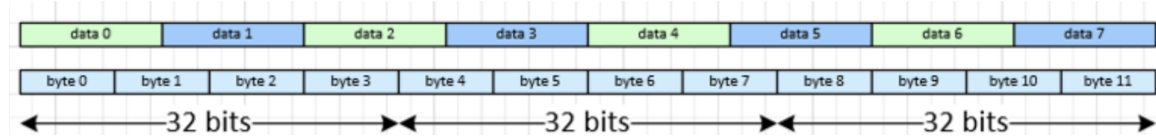


Figure 3.2: Diagram illustrating how data from the two different ADCs is packed in the NAND flash. Each event generates 88,064 bits of waveform data.

3.1.2 Transmit Mode

Once the NAND flash has been populated with data from all 6 signals, it is ready for transmission. There are additional post-trigger blocks based on the sampling rate of the corresponding ADCs, as pictured in Figure 3.3.

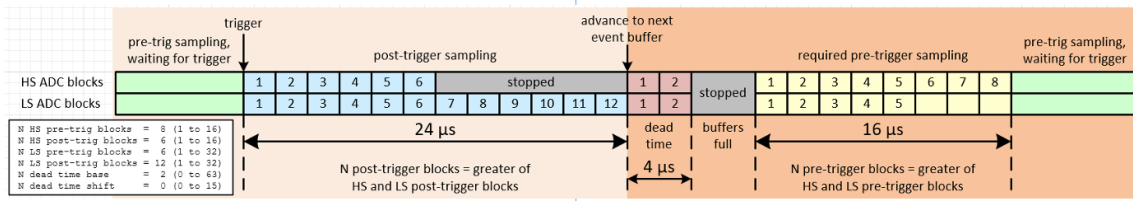


Figure 3.3: Block diagram of the instrument timeline, based on high (HS) and low (LS) sampling rate ADCs.

Once the instrument triggers, there is a $4 \mu\text{s}$ dead time after the required post-trigger sampling has been completed. Hence, at the very least, there will be a $4 \mu\text{s}$ gap between events.

Transitions between instrument modes are driven by commands from the spacecraft. For the purposes of error detection and correction (EDAC), the 30 most recently executed commands and their associated timestamps are stored in the instrument log. This instrument log can be telemetered to the ground with the **DumpLog** command. These 30 commands include successful and failing command executions. However, if the command is so badly mangled that the FPGA or FSW is unable to recognize it as a command, then it does not get included in the log.

3.1.3 Spacecraft Time Interface

Each event is organized chronologically by the FPGA trigger time. For each event, the trigger time is stored in the event header. The event header is always specified by **SCI0TYPE=1** (see 3.3.1 for more details). This is given by a linear combination of the **SHFINE** and **SHCOARSE** variables. For each FPGA header, the **SHCOARSE** variable is a 16-bit unsigned integer that delineates the number of seconds since epoch (Midnight, January 1st, 2012). **SHFINE** variable indicates a number of 20 microsecond intervals in addition to the seconds since epoch. Together, these time measurements establish when an event took place with respect to epoch. The formula to retrieve the epoch from the FPGA header is given by

$$\tau_{epoch} = SHFINE + 20 * 10^{-6} * SHCOARSE \quad (3.1)$$

Additionally, the time axis for each of IDEX's 6 signals is derived from the sampling rate and the number of samples. The Ion Grid and two Target waveforms sample at 4 MHz, and the TOF sample at 130 MHz. The number of samples can then be multiplied by $3.846 [ns]$ and $246.15 [ns]$ for the time axis. **These samples are packed into blocks of 512 samples for the high sampling (HS) rate data and 8 for the low sampling (LS) rate data. The maximum amount of high-rate data is 16 blocks, which would be 8192 samples covering 31.5 us of time. The maximum amount of low-rate data is 64 blocks, which would be 512 samples covering 126.03 us of time.**

To line up the data, you need to line up the trigger point based on the settings for pre-trigger blocks and post-trigger blocks for the high-rate and low-rate data. For this, we turn to the **NBlock-s/TXHDRBLOCKS** variable in each FPGA header. This variable is interpreted according to the map in figure 3.4.

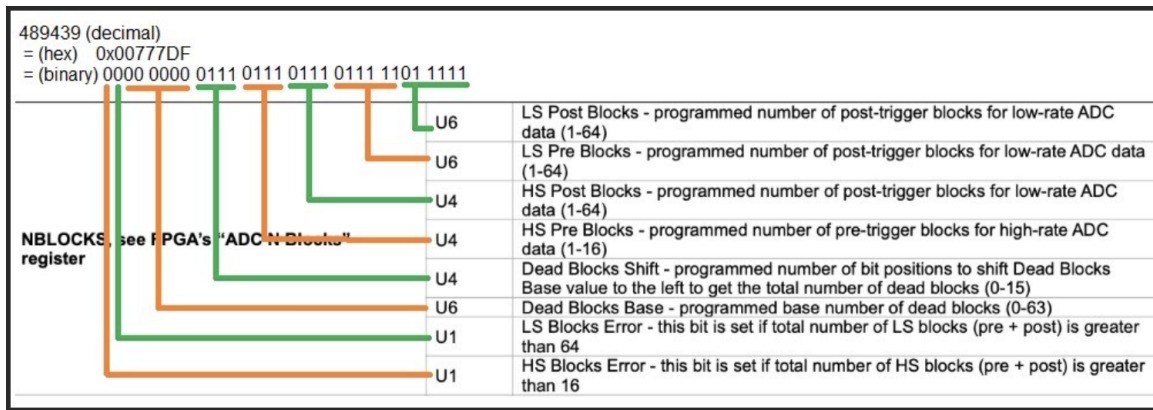


Figure 3.4: Map showing how to decompose the **NBlocks** variable, provided by Greg Newcomb.

The example shown is for a given integer value used during early integration and test, where the trigger was always in the middle of data acquisition (same number of pre and post trigger samples). Generally, this variable is a 32 digit binary string, which can be broken down as follows:

- Digit 1** = 1 or 0 if there is an issue with the number of HS blocks.
- Digit 2** = 1 or 0 if there is an issue with the number of LS blocks.
- Digits 3-8** = Programmed base number of dead blocks: after post trigger sampling?
- Digits 9-12** = Dead blocks shift: Used with 3-8 to determine total number of dead blocks.
- Digits 13-16** = Number of high sampling pre-trigger blocks.
- Digits 17-20** = Number of high sampling post-trigger blocks.
- Digits 21-26** = Number of low sampling pre-trigger blocks.
- Digits 27-32** = Number of low sampling post-trigger blocks.

3.2 Nominal Science Operations

IDEX is an event-driven instrument, so data collection only takes place in the event of a trigger. Otherwise, when in science mode the instrument will continuously fill and delete the memory buffer.

3.3 Data Products

The IDEX science data products pipeline takes raw instrument CCSDS packets (level 0 data) as inputs and goes through each stage of the mass spectrum calibration process to reveal the composition, target impact charge, and ion-grid charge of each measured dust particle (level 3 products). Each science data product level is delineated below in Table 3.1.

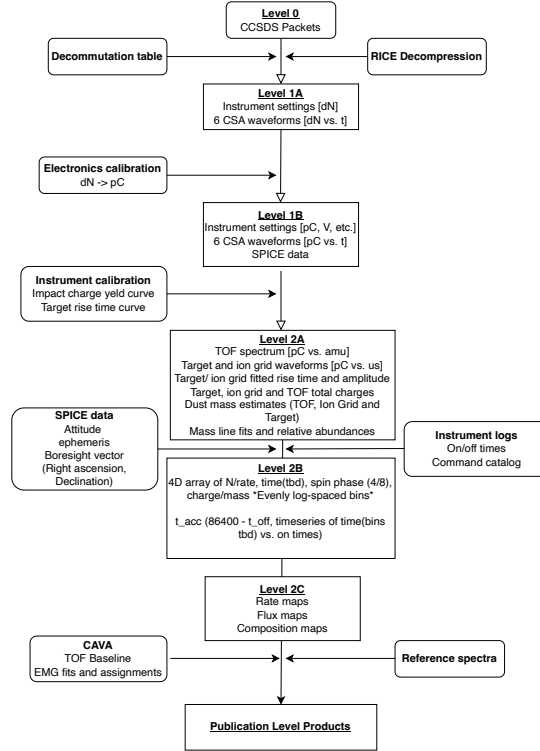


Table 3.1: An outline of the IDEX instrument science data product levels, their format, and a brief description.

3.3.1 Science Data

IDEX generates three APIDs containing science data:

- **Transmit (APID 0x590):** Waveforms and metadata for science events, in response to the **Transmit** command.
- **Fetch (APID 0x591):** Waveforms and metadata for science events, in response to the **Fetch** and **FetchOne** commands.

The contents of each APID are further described below. All IDEX packets are 32-bit aligned, starting with a CCSDS header. only APID 0x590 is expected to be used by IDEX. All IDEX packets are 32-bit aligned, start with a CCSDS header, and end with a 2-byte CRC. Each IDEX packet is encapsulated in an interface transfer frame (ITF) required by the IMAP General Instrument to Spacecraft Interface Control Document (GI ICD, APL #7516-9055), but is ignored in this document.

In general, science data is organized onboard by the **accountability identifier (AID)**. The AID is assigned when the data is acquired, and is used to access the data afterwards. The AID cannot be changed, and duplicate AIDs are not permitted. Each dataset is composed of one or more “**events**”, which is a collection of data collected when the instrument triggers, usually representing a dust particle impact. Each event is given a unique “event number” by the FPGA, so that every event collected by IDEX can be uniquely identified by the combination of AID and event number.

Event data is composed of seven distinct parts:

1. Time of Flight (TOF) waveform data on the high-gain (HG) channel, sampled at 260 megasamples per second (Mps) with 10-bit resolution.
2. TOF waveform data on the mid-gain (MG) channel, sampled at 260 Mps with 10 bit resolution.
3. TOF waveform data on the low-gain (LG) channel, sampled at 260 Mps with 10 bit resolution.
4. Ion (ION) grid Charge Sensitive Amplifier (CSA) waveform data, sampled at 4.0625 Mps with 12 bit resolution.
5. Target Low Range (TLR, AKA target high gain) charge sensitive amplifier (CSA) waveform data, sampled at 4.0625 Mps with 12 bit resolution
6. Target High Range (THR, AKA target low gain) charge sensitive amplifier (CSA) waveform data, sampled at 4.0625 Mps with 12 bit resolution
7. FPGA Pre-pended metadata header containing IDEX’s configuration and status at the time of the trigger, recorded by the FPGA.

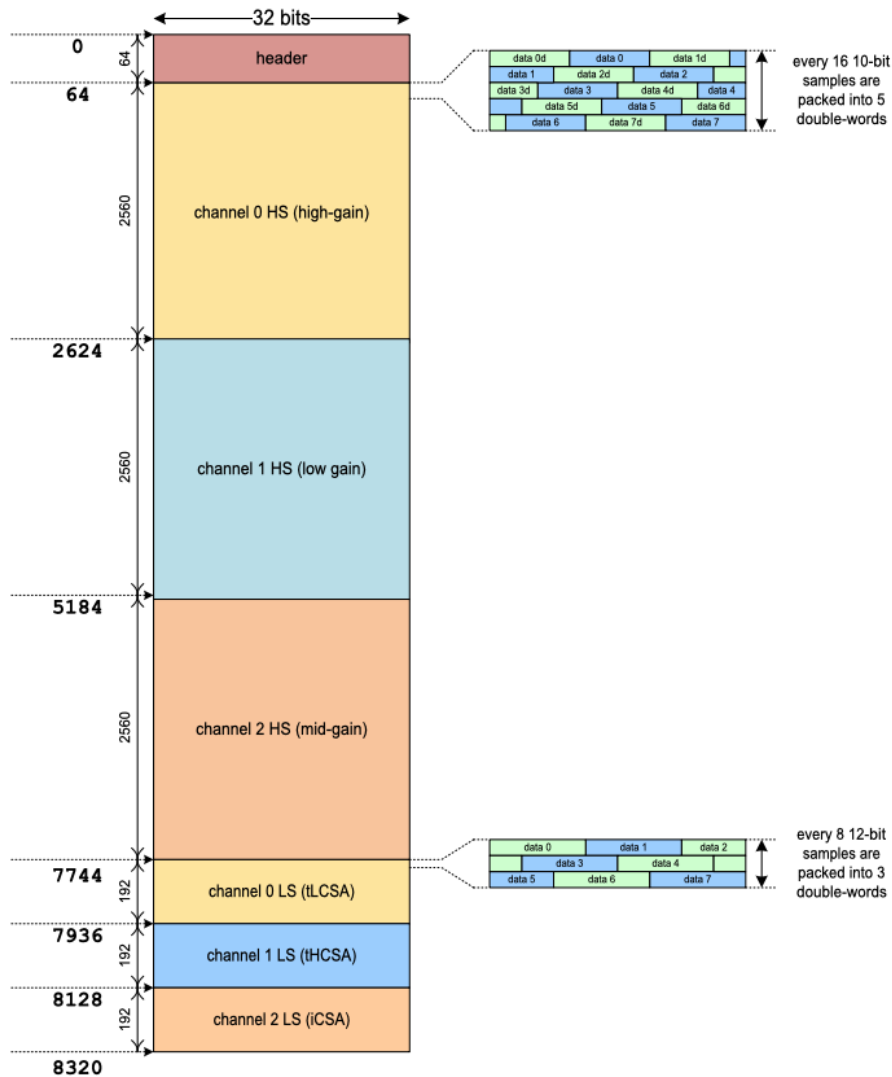


Figure 3.5: A bitmap outline of the science data constituting each event

At the end of science data acquisition, a catalog representing the new dataset is stored along with the raw science data in nonvolatile memory. The catalog is mostly empty at this point but can be downlinked with the **SendCatalog** command. If no science data is acquired during acquisition, then no catalog is created either. A high-level diagram of how the high and low sampling rate ADC data is transmitted once packetized is shown in Figure 3.5, describing how different blocks of data are stored.

Following science data acquisition, FSW will transmit the science data with the “Process_Transmit” command using APID 0x590. Processing is started with the **Process** command, which has the FSW evaluating waveform data to determine characteristics (e.g. peak count) of each event. The characteristics are then used to assign each individual event to a category. As each event in a dataset

is processed, a new entry is added to the catalog (in chronological order of acquisition). The FSW includes a setting where event data (all seven parts) are transmitted (APID 0x590) as soon as it is processed – which was used for some early ground testing but never afterward.

A dataset only needs to be processed once but can be processed multiple times (with different results if parameter table entries are changed). However only the most recent processing results are saved to the catalog.

Once a dataset is processed, it can be downlinked as any of the three packet types. Normally the catalog is sent right away since it is considered “feed-forward data” used to configure instrument parameters for subsequent data acquisition opportunities. “Transmitted” science data comes next, and are used as a “first pass” at evaluating the science data and downlinking a subset of acquired events. Afterward, scientists will evaluate the catalog and request specific events to be downlinked as “fetched” science data. Another option is that the dataset can be “transmitted” again but with different parameter table entries resulting in a different selection of events transmitted.

Transmitted and Fetched Science Packets

Transmitted and fetched science data share the same decommutation, the only real difference being the APID. When transmitting or fetching science data, one event is handled at a time. Each event may spawn between 1 and 10 packets, depending on size and requested contents.

As mentioned above, event data is composed of seven distinct parts but not all are sent for every event. An argument of the **FetchOne** command specifies which parts to send, and similarly the Fetch Table includes this info for “Fetch” commands (see **AddToFetch** command for more details). Parameter table entries are used to determine what parts of an event are sent for transmitted data (using the event’s category which themselves are grouped into quality factors). Each part of the event is packaged into its own packet, while still sharing the same APID. To fit within the IICD-required maximum CCSDS packet length of 4095 bytes, one packet will contain no more than 4032 bytes of science data (the rest dedicated to headers and such). Each part normally fits in one packet, but the TOF waveforms can span two-three packets if the dataset is large and compression is poor (or disabled). In this case, the maximum amount of science data is sent in the first packet and any leftover sent in the next packet. The content of packets containing the metadata header are very different from the waveform data, so it is discussed separately.

Packets Containing Waveforms

Science packets containing any of the six waveforms, whether transmitted or fetched, start with a brief header and are followed by a variable amount of science data. The header is never compressed, and provides identifying information such as AID, event number, and which waveform.

Item Name	Type	Description
VERSION	U3	CCSDS version number, fixed 0b000
TYPE	U1	CCSDS type, fixed 0b0 for telemetry
SEC_HDR_FLG	U1	CCSDS secondary header flag, fixed 0b1 for presence of secondary header
PKT_APID	U11	CCSDS Application Process Identifier (APID), either 0x590 or 0x591
SEQ_FLGS	U2	CCSDS grouping flags, fixed 0xb11 for standalone packet
SRC_SEQ_CTR	U14	CCSDS source sequence counter, increments per packet (separate counters per APID)
PKT_LEN	U16	CCSDS packet length, excludes UMS and CCSDS primary header (minus one)
SHCOARSE	U32	Time of packet generation (not data acquisition), as integer seconds since epoch
SHFINE	U16	Time of packet generation, each DN represents 20usec within current second
SCI0AID	U32	Accountability Identifier for this event
SCI0TYPE	U8	Packet data content type: 0x01: FPGA pre-pended metadata header 0x02: TOF HG 0x04: TOF LG 0x08: TOF MG 0x10: Target High Range (THR) CSA 0x20: Target Low Range (TLR) CSA 0x40: Ion grid (ION) CSA
SCI0CONT	U8	Channels being downlinked (same decomposition as above)
SCI0SPARE1	U13	Spare
SCI0PACK	U1	Always 0b1, showing data is bit-packed
SCI0FRAG	U1	If 0b1, then this channel data spans multiple packets and this is not the last packet (use SCI0FRAGOFF to reconstruct)
SCI0COMP	U1	Data is compressed if 0b1, otherwise not
SCI0EVNUM	U16	Event number for this event
SCI0CAT	U8	Category assigned to this event (for most recent processing operation)
SCI0QUAL	U8	Quality factor assigned to this category
SCI0FRAGOFF	U16	Starting offset for this data when reconstructing data that spans multiple packets, in 32-bit "dwords" not bytes. Either 0x000 or multiple of 0x3F0
SCI0VER	U16	Science CSC version number for this header
SCI0TIMECOARSE	U16	Lowest 16 bits of integer seconds since epoch for time of this event
SCI0TIMEFINE	U16	Sub-second time of first event, each DN represents 20usec within current second
SCI0SPARE2	U32	Spare
SCI0SPARE3	U32	Spare
SCI0SPARE4	U32	Spare
Variable length raw science data	Varies	Padded with zeroes for 32-bit alignment, if needed
SYNCSOI0PKT	U16	Synchronization marker, fixed 0x3333
CRCSCI0PKT	U16	CRC CCITT-16

Outline of fields in waveform-containing packets. Red text indicates changes from the SUDA definitions.

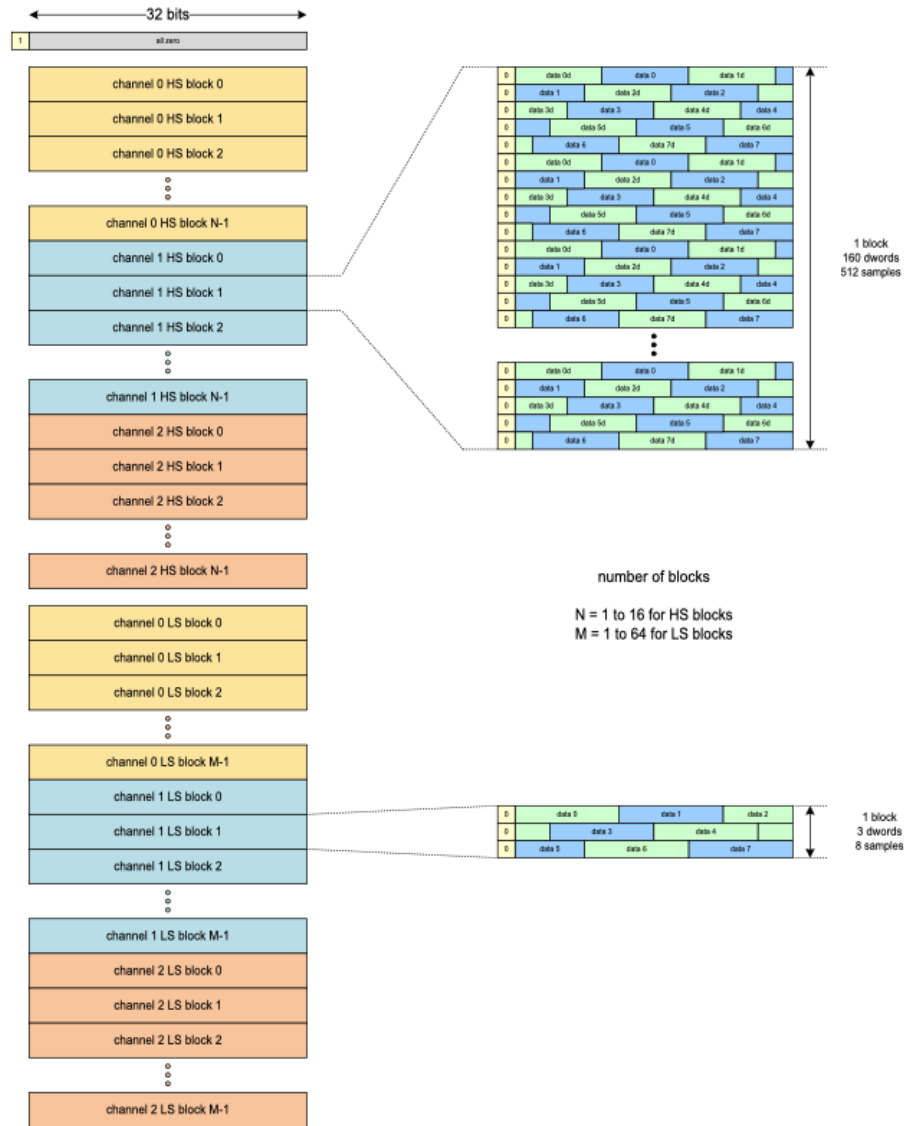


Figure 3.6: A bitmap outline of what is contained in each set of waveform data.

The FSW can make use of the FPGA's RICE compression core, with details discussed in the FPGA specification (167130). **If the resulting compressed data is not 32-bit aligned, then the remainder (up to 31 bits) are padded with zeroes.** To decompress a TOF waveform, a RICE decompressor should be used and the output interpreted as a chronological series of 10-bit samples. For SCA waveforms, the output is 12-bit samples.

After the data is decompressed on the ground, the resulting data represents 6 time-ordered waveforms

corresponding to each AID.

The packet is summarized in the table below, using a format similar to the SUDA Command and Telemetry handbook:

- Item name: Mnemonic for the named item
- Bytes: Position of the item within the element, show bytes with packet and bits within byte
- Type of data: Everything is “U” for unsigned data number, followed by bit length
- Description: Briefly describes the item, giving the conversion if appropriate.

The **SCI0RAW** field contains the instrument waveforms in each packet. How the header, low sampling and high sampling rate ADC data appears in each packet is described in figure 3.6. While each channel appears in 3 packets for most events, figure 3.6 describes the general case for N high sampling and M low sampling blocks. Each packet header, on the other hand, is only contained in one packet.

Packets Containing FPGA Metadata Headers

The packet containing the metadata header holds status and configuration information at the time when this event was acquired. This includes the voltages, currents, temperatures and other instrument settings at trigger time. The first 32 bytes after the CCSDS header are identical to the preceding waveform data, so that the SCI0TYPE field can be used as a key to differentiate packets containing this metadata header versus waveforms.

After the common header is the **FPGA prepended metadata header** that the FPGA captures for every event. These data are fixed size and never compressed. Many of these fields are the same for all events within the same dataset. A diagram describing the components of the FPGA header is shown in Figure 3.7.

packet length in dwords (31:0)																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																			
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Item Name	Type	Description
CCSDS header	12 bytes	See section 2.1 for details
SCI0AID	U32	See above (same as waveform packets)
SCI0TYPE	U8	
SCI0CONT	U8	
SCI0SPARE1	U13	
SCI0PACK	U1	
SCI0FRAG	U1	
SCI0COMP	U1	
SCI0EVTNUM	U16	
SCI0CAT	U8	
SCI0QUAL	U8	
SCI0FRAGOFF	U16	
SCI0VER	U16	
SCI0TIMECOARSE	U16	
SCI0TIMEFINE	U16	
SCI0SPARE2	U32	
SCI0SPARE3	U32	
SCI0SPARE4	U32	
SCIDATALENGTH	U32	Length of raw science data in memory including this FPGA pre-pended metadata header but excludes the sync words at the start of all Flash pages.
TIMESTAMP1	U16	Padded with 0's
	U16	Time of event trigger, in integer seconds (bits 31:16)
TIMESTAMP2	U16	Time of event trigger, in integer seconds (bits 15:0)
	U16	Time of event trigger, in sub-seconds (20usec per DN) event time, sub-seconds
EVENTNUMBER	U1	Padded with 0
	U3	Trigger offset: number of WCLK cycles that the start of post-trigger data collection is delayed from the trigger position to align the HS and LS sampling periods. See FPGA's "ADC Trigger Offset" register.
	U2	Padded with 0's
	U10	Identifies origin of this trigger (multiple concurrent triggers are possible)
		· upper 4 bits are spares (unused)
		· bit 5 – external trigger
		· bit 4 – SW trigger
		· bit 3 – LS ADC1 trigger (target low range CSA)
		· bit 2 – HS ADC1Q trigger (mid-gain channel)
		· bit 1 – HS ADC0Q trigger (low-gain channel)
		· bit 0 – HS ADC0I trigger (high-gain channel)
		See FPGA's "ADC Trigger ID" register.
	U16	Event number assigned by FPGA. Resets to 0 at power on and incremented thereafter (rolling over from 0xFFFF to 0).
NBLOCKS, see FPGA's "ADC N Blocks" register	U1	HS Blocks Error - this bit is set if total number of HS blocks (pre + post) is greater than 16
	U1	LS Blocks Error - this bit is set if total number of LS blocks (pre + post) is greater than 64
	U6	Dead Blocks Base - programmed base number of dead blocks (0-63)
	U4	Dead Blocks Shift - programmed number of bit positions to shift Dead Blocks Base value to the left to get the total number of dead blocks (0-15)
	U4	HS Pre Blocks - programmed number of pre-trigger blocks for high-rate ADC data (1-16)
	U4	HS Post Blocks - programmed number of post-trigger blocks for low-rate ADC data (1-64)
	U6	LS Pre Blocks - programmed number of post-trigger blocks for low-rate ADC data (1-64)
	U6	LS Post Blocks - programmed number of post-trigger blocks for low-rate ADC data (1-64)
HSADC0IHDR1, 000	U10	ADC0I trigger level - trigger level for ADC0I channel

HSADC0IHDR1, see FPGA's "HS ADC0 Channel I Trigger Control 2" register	U11	ADC0I trigger Nmax12 - max # of samples between pulse 1 and 2 for ADC0I double pulse triggering
	U11	ADC0I trigger Nmin12 - min # of samples between pulse 1 and 2 for ADC0I double pulse triggering
HSADC0IHDR2, see FPGA's "HS ADC0 Channel I Trigger Control 1" register	U8	ADC0I trigger Nmax2 - max # of samples for pulse 2 for ADC0I double pulse mode triggering
	U8	ADC0I trigger Nmin2 - min # of samples for pulse 2 for ADC0I double pulse mode triggering
	U8	ADC0I trigger Nmax1 - max # of samples for pulse 1 for ADC0I single and double pulse triggering
	U8	ADC0I trigger Nmin1 - min # of samples for pulse 1 for ADC0I single and double pulse triggering
HSADC0QHDR1, see FPGA's "HS ADC0 Channel Q Trigger Control 2" register	U10	ADC0Q trigger level - trigger level for ADC0Q channel
	U11	ADC0Q trigger Nmax12 - max # of samples between pulse 1 and 2 for ADC0Q double pulse triggering
	U11	ADC0Q trigger Nmin12 - min # of samples between pulse 1 and 2 for ADC0Q double pulse triggering
HSADC0QHDR2, see FPGA's "HS ADC0 Channel Q Trigger Control 1" register	U8	ADC0Q trigger Nmax2 - max # of samples for pulse 2 for ADC0Q double pulse mode triggering
	U8	ADC0Q trigger Nmin2 - min # of samples for pulse 2 for ADC0Q double pulse mode triggering
	U8	ADC0Q trigger Nmax1 - max # of samples for pulse 1 for ADC0Q single and double pulse triggering
	U8	ADC0Q trigger Nmin1 - min # of samples for pulse 1 for ADC0Q single and double pulse triggering
HSADC1QHDR1, see FPGA's "HS ADC1 Channel Q Trigger Control 2" register	U10	ADC1Q trigger level - trigger level for ADC1Q channel
	U11	ADC1Q trigger Nmax12 - max # of samples between pulse 1 and 2 for ADC1Q double pulse triggering
	U11	ADC1Q trigger Nmin12 - min # of samples between pulse 1 and 2 for ADC1Q double pulse triggering
HSADC1QHDR2, see FPGA's "HS ADC1 Channel Q Trigger Control 1" register	U8	ADC1Q trigger Nmax2 - max # of samples for pulse 2 for ADC1Q double pulse mode triggering
	U8	ADC1Q trigger Nmin2 - min # of samples for pulse 2 for ADC1Q double pulse mode triggering
	U8	ADC1Q trigger Nmax1 - max # of samples for pulse 1 for ADC1Q single and double pulse triggering
	U8	ADC1Q trigger Nmin1 - min # of samples for pulse 1 for ADC1Q single and double pulse triggering
LSADC, see FPGA's LS ADC1 Trigger Control" register	U8	Padded with 0's
	U3	coincidence mode blocks - number of blocks coincidence window is enabled after LS ADC1 trigger
	U1	LS ADC1 trigger polarity - trigger polarity for LS ADC1 (0 = normal, 1 = inverted)
	U12	LS ADC1 trigger level - trigger level for LS ADC1
	U8	LS ADC1 trigger Nmin - min number of samples above/below trigger level for triggering LS ADC1
TRIGGERMODE, see FPGA's "Trigger Mode" Register	U1	HVPS polarity status, "cation" mode (0) or "anion" mode (1)
	U1	Stim pulse enabled (1) or disabled (0)
	U21	Padded with 0's
	U1	External trigger polarity
	U1	LS ADC coincidence mode enable
	U1	LS ADC trigger enable
	U2	ADC1Q trigger mode – same as above
	U2	ADC0Q trigger mode – same as above
	U2	ADC0I trigger mode 00 – no triggering 01 – threshold mode 10 – single pulse mode 11 – double pulse mode
	U2	
SPARE0	U32	Padded with 0's
SPARE1	U32	Padded with 0's
	U20	Padded with 0's

IONDETAILS0	U12	Ion grid (LS ADC2) 15 ms min, new for IDEX. The minimum value on LS ADC2 during the 15 ms window after the previous event trigger.
IONDETAILS1	U16	Ion grid (LS ADC2) saturation high count, new for IDEX. The number of LS ADC2 samples that were equal to 0x000 during the 15 ms window after the previous event trigger.
	U16	Ion grid (LS ADC2) saturation low count, new for IDEX. The number of LS ADC2 samples that were equal to 0xFF during the 15 ms window after the previous event trigger.
TOFMAX	U2	Padded with 0's
	U10	ADC1Q max value - max sample value on ADC1Q channel since previous event was captured. See FPGA's "ADC1 DQ Channel MinMax" Register.
	U10	ADC0Q max value - max sample value on ADC0Q channel since previous event was captured. See FPGA's "ADC0 DQ Channel MinMax" Register.
	U10	ADC0I max value - max sample value on ADC0I channel since previous event was captured. See FPGA's "ADC0 DI Channel MinMax" Register.
TOFMIN	U2	Padded with 0's
	U10	ADC1Q min value - min sample value on ADC1Q channel since previous event was captured. See FPGA's "ADC1 DQ Channel MinMax" Register.
	U10	ADC0Q min value - min sample value on ADC0Q channel since previous event was captured. See FPGA's "ADC0 DQ Channel MinMax" Register.
	U10	ADC0I min value - min sample value on ADC0I channel since previous event was captured. See FPGA's "ADC0 DI Channel MinMax" Register.
LSADC0MINMAX, see FPGA's "LS ADC0 Channel MinMax" Register	U8	Padded with 0's
	U12	LS ADC0 max value - max sample value on LS ADC0 channel since previous event was captured
	U12	LS ADC0 min value - min sample value on LS ADC0 channel since previous event was captured
LSADC1MINMAX, see FPGA's "LS ADC1 Channel MinMax" Register	U8	Padded with 0's
	U12	LS ADC1 max value - max sample value on LS ADC1 channel since previous event was captured
	U12	LS ADC1 min value - min sample value on LS ADC1 channel since previous event was captured
LSADC2MINMAX, see FPGA's "LS ADC2 Channel MinMax" Register	U8	Padded with 0's
	U12	LS ADC2 max value - max sample value on LS ADC2 channel since previous event was captured
	U12	LS ADC2 min value - min sample value on LS ADC2 channel since previous event was captured
TOFTRIGGERDELAY	U2	Padded with 0's
	U10	ADC1Q trigger delay - delay in DCLK of trigger condition through comparison FIFO for ADC1Q
	U10	ADC0Q trigger delay - delay in DCLK of trigger condition through comparison FIFO for ADC0Q
	U10	ADC0I trigger delay - delay in DCLK of trigger condition through comparison FIFO for ADC0I
TOFSAMPLEDELAY	U2	Padded with 0's
	U10	ADC1Q sample delay - delay in samples from actual trigger position in event buffer to expected trigger position for ADC1Q
	U10	ADC0Q sample delay - delay in samples from actual trigger position in event buffer to expected trigger position for ADC0Q
	U10	ADC0I sample delay - delay in samples from actual trigger position in event buffer to expected trigger position for ADC0I
TOFTRANSCOUNT, see FPGA's "Transitions Counts" Register	U2	Padded with 0's
	U10	ADC1Q transition count - Number of transitions across the programmed threshold level for ADC1Q
	U10	ADC0Q transition count - Number of transitions across the programmed threshold level for ADC0Q
	U10	ADC0I transition count - Number of transitions across the programmed threshold level for ADC0I
PROCHKADCCHAN01	U4	Padded with 0's
	U12	Last measurement in raw DN for Processor Board signal "current on the 1V POL"
	U4	Padded with 0's

	U12	Last measurement in raw DN for Processor Board signal "current on the 1.9V POL"
PROCHKADCCHAN23	U4	Padded with 0's
	U12	Last measurement in raw DN for Processor Board signal "ProcBd Temperature 1" (near TOF/HS ADCs)
	U4	Padded with 0's
	U12	Last measurement in raw DN for Processor Board signal "ProcBd Temperature 2" (near PWB center)
PROCHKADCCHAN45	U4	Padded with 0's
	U12	Last measurement in raw DN for Processor Board signal "voltage on 1V bus"
	U4	Padded with 0's
	U12	Last measurement in raw DN for Processor Board signal "FPGA Temperature"
PROCHKADCCHAN67	U4	Padded with 0's
	U12	Last measurement in raw DN for Processor Board signal "voltage on 1.9V bus"
	U4	Padded with 0's
	U12	Last measurement in raw DN for Processor Board signal "voltage on 3.3V bus"
HVPSHKADCCHAN01	U4	Padded with 0's
	U12	Last measurement in raw DN for HVPS Board signal "Detector Voltage"
	U4	Padded with 0's
	U12	Last measurement in raw DN for HVPS Board signal "Sensor Voltage"
HVPSHKADCCHAN23	U4	Padded with 0's
	U12	Last measurement in raw DN for HVPS Board signal "Target Voltage"
	U4	Padded with 0's
	U12	Last measurement in raw DN for HVPS Board signal "Reflectron Voltage"
HVPSHKADCCHAN45	U4	Padded with 0's
	U12	Last measurement in raw DN for HVPS Board signal "Rejection Voltage"
	U4	Padded with 0's
	U12	Last measurement in raw DN for HVPS Board signal "current for the HVPS detector"
HVPSHKADCCHAN67	U4	Padded with 0's
	U12	Last measurement in raw DN for HVPS Board signal "positive current for the HVPS sensor"
	U4	Padded with 0's
	U12	Last measurement in raw DN for HVPS Board signal "negative current for the HVPS sensor"
LVPSHKADC0CHAN01	U4	Padded with 0's
	U12	Last measurement in raw DN for LVPS Board signal "voltage of +3.3V reference"
	U4	Padded with 0's
	U12	Last measurement in raw DN for LVPS Board signal "voltage of +3.3V operational reference"
LVPSHKADC0CHAN23	U4	Padded with 0's
	U12	Last measurement in raw DN for LVPS Board signal "voltage on -6V bus"
	U4	Padded with 0's
	U12	Last measurement in raw DN for LVPS Board signal "voltage on +6V bus"
LVPSHKADC0CHAN45	U4	Padded with 0's
	U12	Last measurement in raw DN for LVPS Board signal "voltage on +16V bus"
	U4	Padded with 0's
	U12	Last measurement in raw DN for LVPS Board signal "voltage on +3.3V bus"
LVPSHKADC0CHAN67	U4	Padded with 0's
	U12	Last measurement in raw DN for LVPS Board signal "voltage on -5V bus"
	U4	Padded with 0's
	U12	Last measurement in raw DN for LVPS Board signal "voltage on +5V bus"
LVPSHKADC1CHAN01	U4	Padded with 0's
	U12	Last measurement in raw DN for LVPS Board signal "current on +3.3V bus"
	U4	Padded with 0's

LVPSHKADC1CHAN2 3	U12	Last measurement in raw DN for LVPS Board signal “current on 16V bus”
	U4	Padded with 0’s
	U12	Last measurement in raw DN for LVPS Board signal “current on +6V bus”
	U4	Padded with 0’s
LVPSHKADC1CHAN4 5	U12	Last measurement in raw DN for LVPS Board signal “current on -6V bus”
	U4	Padded with 0’s
	U12	Last measurement in raw DN for LVPS Board signal “current on +5V bus”
	U4	Padded with 0’s
LVPSHKADC1CHAN6 7	U12	Last measurement in raw DN for LVPS Board signal “current on -5V bus”
	U4	Padded with 0’s
	U12	Last measurement in raw DN for LVPS Board signal “current on +2.5V bus”
	U4	Padded with 0’s
LVPSHKADC2CHAN0 1	U12	Last measurement in raw DN for LVPS Board signal “current on -2.5V bus”
	U4	Padded with 0’s
	U12	Last measurement in raw DN for LVPS Board signal “primary target temperature”
	U4	Padded with 0’s
LVPSHKADC2CHAN2 3	U12	Last measurement in raw DN for LVPS Board signal “redundant target temperature”
	U4	Padded with 0’s
	U12	Last measurement in raw DN for LVPS Board signal “temperature of Decon-Actuator Board (DAB)”
	U4	Padded with 0’s
LVPSHKADC2CHAN4 5	U12	Last measurement in raw DN for LVPS Board signal “temperature of LVPS board”
	U4	Padded with 0’s
	U12	Last measurement in raw DN for LVPS Board signal “temperature of HVPS board (in EBox)”
	U4	Padded with 0’s
LVPSHKADC2CHAN6 7	U12	Last measurement in raw DN for LVPS Board signal “voltage on +2.5V bus”
	U4	Padded with 0’s
	U12	Last measurement in raw DN for LVPS Board signal “voltage on -2.5V bus”
	U4	Padded with 0’s
SPARE	U32	Last measurement in raw DN for LVPS Board signal “Spare, which is not connected to anything”
SPARE	U32	Padded with 0’s
FSWHDR00	U32	Padded with 0’s
FSWHDR01	U32	Section reserved for FSW usage
FSWHDR02	U32	Section reserved for FSW usage
FSWHDR03	U32	Section reserved for FSW usage
FSWHDR04	U32	Section reserved for FSW usage
FSWHDR05	U32	Section reserved for FSW usage
FSWHDR06	U32	Section reserved for FSW usage
FSWHDR07	U32	Section reserved for FSW usage
FSWHDR08	U32	Section reserved for FSW usage
FSWHDR09	U32	Section reserved for FSW usage
FSWHDR10	U32	Section reserved for FSW usage
FSWHDR11	U32	Section reserved for FSW usage
FSWHDR12	U32	Section reserved for FSW usage
FSWHDR13	U32	Section reserved for FSW usage
FSWHDR14	U32	Section reserved for FSW usage
FSWHDR15	U32	Section reserved for FSW usage
FPGAVER	U32	Section reserved for FSW usage
SYNCSIOPKT	U32	FPGA version, see FPGA’s “DataTime Register”
CRCSIOPKT	U16	Synchronization marker, fixed 0x3333
	U16	CRC CCITT-16

Chapter 4

Data Processing

This chapter describes the necessary steps to convert data between each of the levels described in the processing flowchart. Source code snippets are provided in Python.

4.1 Overview

The IDEX data processing pipeline takes the data packets described above, an xml XTCE definition/decommutation table ([B](#)), the instrument ground calibration curve, the spacecraft thruster data, and ephemerides data as inputs. Each data product and its corresponding inputs/outputs is described in Figure [4.1](#).

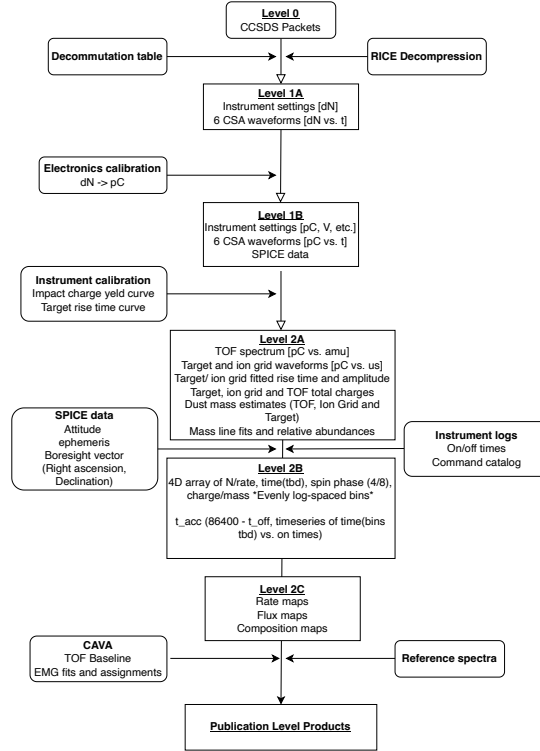


Figure 4.1: The IDEX science data pipeline processing flowchart. Data products are given in the center, with the necessary auxiliary inputs on the sides.

4.2 Data Volume Expectancy

The estimates for the expected IDEX instrument science data volume are derived from simulations of known IDP and ISD populations. Figure 4.2 depicts counts for ISD (red/blue) and IDP (black) particles expected to be measured by IDEX during the mission.

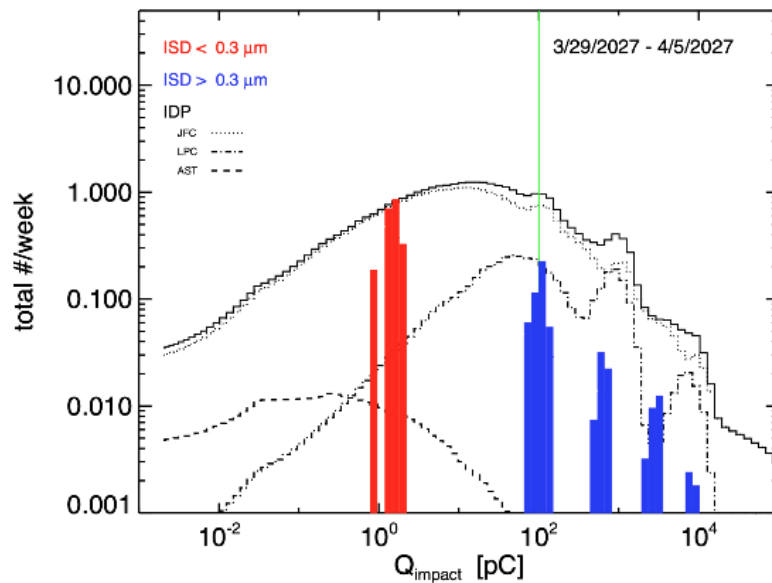


Figure 4.2: Expectations for IDEX’s ISD and IDP measurements during the mission. Note that although this is just shown for 6 days in 2027, these simulations have been carried out for all of the prime and extended IMAP mission.

From this simulation, IDEX is expected to measure 16 events per day on average. Using these estimates in conjunction with the bitmap in Figure 3.7, we can estimate the weekly data volume expectancy for IDEX during the mission. This has been carried out for each of the science data product levels, as depicted in table 4.1.

PRODUCT	PRODUCT STANDARD	DESCRIPTION
LEVEL 0	Packet	288 Kilobits per event * 16 Events per day * 7 events per week = 40,852 Kilobits per week
LEVEL 1A	Raw	40 voltages, currents and temperatures tracked in DN (40*12 = 1,280 bits), 264,192 bits for all 6 signals, so 265,472 per event * 7 * 16 = 29,733 Kilobits per week
LEVEL 1B	Raw	Conversion to Voltage does not introduce new volume, 29,733 Kilobits per week
LEVEL 1C	Reduced	2 double words for each charge + 81,920 bits for TOF spectra = 81,984 bits per event * 7 * 16 = 9,183 Kilobits per week
LEVEL 2	Calibrated	1 Voltage array + 1 double word for mass + up to 12 double words for mass index = 82,336 bits per event * 7 * 16 = 9,222 Kilobits per week
LEVEL 3	Derived	Up to 12 mass lines and their relative abundances = 24 double words = 768 bits per event * 7 * 16 = 87 Kilobits per week
CALIBRATION	Ancillary	Library of laboratory mass spectra of candidate minerals, SPICE kernels for spacecraft state vector and attitude

Table 4.1: Expectations for IDEX’s weekly data volume throughout the mission, organized by data product level.

4.3 Heritage Data Processing

IDEX has a high-level of heritage from the SURface Dust Analyzer (SUDA) onboard the upcoming flagship *Europa Clipper* mission. SUDA is nearly identical but has instead been optimized for flybys and detections of organic molecules for a limited impact velocity range of around 4 km/s. The XTCE definition is the same for both instruments (see appendix B), but the interpretation of the waveforms differs, along with voltage/temperature conversions.

Like IDEX, SUDA Event data is composed of seven distinct parts:

1. Time of Flight (TOF) waveform data on the high-gain (HG) channel, sampled at 250 megasamples per second (Msps) with 10 bit resolution
2. TOF waveform data on the mid-gain (MG) channel, sampled at 250 Msps with 10 bit resolution
3. TOF waveform data on the low-gain (LG) channel, sampled at 250 Msps with 10 bit resolution
4. Velocity (Qv) grid charge sensitive amplifier (CSA) waveform data, sampled at 3.75 Msps with 12 bit resolution
5. Ion (Qi) grid CSA waveform data, sampled at 3.75 Msps with 12 bit resolution
6. Target (Qt) CSA waveform data, sampled at 3.75 Msps with 12 bit resolution
7. FPGA Pre-pended metadata header containing IDEX's configuration and status at the time of the trigger, recorded by the FPGA

The velocity grid CSA takes the place of IDEX's second target gain stage. Otherwise, the names, sampling rates, and bit resolutions are all identical. SUDA leverages the use of the SWIFT language-based SPECTRUM software, developed in-house at LASP. To avoid this Mac OS dependency, the IDEX pipeline has been constructed exclusively in Python 3.8.

4.4 Level 0 to Level 1A Processing

4.4.1 Decommulation

Packet decommutation relies on the **LASP packets** library, available on the [LASP Bitbucket \(LASP VPN access required\)](#). The **LASP packets** software hinges on the **generator** method of the **Parsing.PacketParser** module for reading CCSDS packets (please see appendix A for Python source code).

The packet data is read in using Python's **bitstring** library to reduce memory usage. Each packet is associated with a "flattened sequence container" from the XTCE definition. To do this, first, the length of the packet is taken from the CCSDS header:

```
[breaklines, xleftmargin=20pt]python specified_packet_data_length = self.total_packet_bits_from_pktlen(header['PKTLEN'])
```

Next, the type of packet as provided by the CCSDS header is cross-referenced with the XTCE definition to find a matching APID, VERSION, and TYPE. If there is a match, the code parses the packet according to the XTCE definition. If there is no match, it will attempt to generally parse it using the length of the packet. The source for this is shown below.

```
[breaklines, xleftmargin=20pt]python try: if isinstance(self.packet_definition, xtcedef.XtcePacketDefinition):
packet = self.parse_packet(binary_data, self.packet_definition.named_containers, root_container_name =
root_container_name, word_size = self.word_size)else : packet_definition, parameter_list = self.determine_packet_restrictions
self.legacy_parse_packet(binary_data, parameter_list, word_size = self.word_size)
```

This process is repeated until we have parsed each of the packets in the dataset.

4.4.2 Science Data

Once the data has been depacketized, the **SCI0TYPE** field designates what data is contained in each packet. Per the table in [3.3.1](#), the possible values of **SCI0TYPE** include:

1 = FPGA pre-pended metadata header
2 = TOF HG
4 = TOF LG
8 = TOF MG
10 = Target LG (Q_T) CSA
20 = Target HG (Q_T) CSA
40 = Ion grid (Q_I) CSA

By reading in the **SCI0TYPE** field, the appropriate ADC sampling rate (and therefore bit pattern) is determined.

The waveform data must be read in a pattern specific to high or low sampling rate ADCs in the schema outline below:

For **high sampling rate (TOF) data (10-bit samples)**, handle the raw data as 32-bit units like this (most significant bit first):

2 bits = Padding set to zeros, ignore this
10 bits = oldest sample
10 bits = next sample
10 bits = next (newest) sample

For the remaining **low sampling rate data (12-bit samples)**, handle the raw data as 32-bit units like this (most significant bit first):

8 bits = Padding set to zeros, ignore this
12 bits = oldest sample
12 bits = next (newest) sample

Once this schema has been used for the waveform data, all the science event data has been read in and stored. The example Python script below will read in an IDEX raw binary data file, plot each of the event waveforms, and write all of the data to an easily accessible HDF5 file. Plots are saved to a */Plots/* directory in the same directory as the script, and the HDF5 files are written to the */HDF5/* directory.

It has been updated to be used as a command line tool, so we may use: [breaklines]python python *idx_packet.py* *--filefilename* Where *filename* must be replaced with the name of the packet file, and *python* must be an alias for a python installation. The *--file* can be substituted for *-f*.

4.4.3 Housekeeping data

The IDEX instrument settings delineated in table **** are all decommutated and stored with their raw DN values in the CDF file for level 1A.

4.5 Level 1A to Level 1B Processing

Level 1B data contains both instrument settings and waveforms in physical units, rather than in dN. For the waveforms, a linear transformation is used to convert to pC. These values have been established for the IDEX FM, and are tabulated in 4.2.

A polynomial transformation is used to convert the instrument settings from dN to physical units. Below is a table of all the instrument settings and their polynomial conversion values.

Here, (C0,C1,...) are the coefficients for the polynomial transformation. The equation to transform from dN to physical units is then

$$P(dN) = \sum_0^i C_i(dN)^i, \quad (4.1)$$

where P is the value in physical units and dN represents the value in dN.

Table 4.2: IDEX Flight Gains, updated on 10/10/24.

	Variable (pC/DN)	Conv. Factor (pC/DN)	Sampling Rate (MHz)	Bits
High Gain TOF	$Q_{tof,tot}$	2.89×10^{-4}	260	10
Mid Gain TOF	$Q_{tof,tot}$	1.13×10^{-2}	260	10
Low Gain TOF	$Q_{tof,tot}$	5.14×10^{-1}	260	10
Ion Grid	Q_{iG}	7.46×10^{-4}	4.0625	12
Target High Gain	Q_{tgt}	1.63×10^{-1}	4.0625	12
Target Low Gain	Q_{tgt}	1.58×10^1	4.0625	12

4.6 Level 1B to Level 2A Processing

The transition from 1B to 2A involves performing fits to each of IDEX's signals, then deriving quantities from those fit parameters to estimate the mass, speed and composition of the impacting dust particle.

Target and Ion Grid Signal Fits

The level 1B to 2A conversion uses fits to the ion grid and two target signals to estimate the total impact charge generated by the dust particle. The mass and speed are derived from these fit parameters and our calibration data.

The fit function is given by (Horányi, 2014),

$$y(t; t_0, C_0, C_1, C_2, \tau_0, \tau_1, \tau_2) = C_0 + H(t - t_0)[C_2(1 - e^{-\frac{t-t_0}{\tau_1}})e^{-\frac{t-t_0}{\tau_2}} - C_1] \quad (4.2)$$

where $H(t - t_0)$ is the Heaviside function. Here, C_0 is the initial baseline.

The impact charge is calculated by taking the difference between the initial baseline and the peak of the signal. An example fit for the ion grid signal is depicted in figure 4.3.

Ion Grid Signal Fit number 8

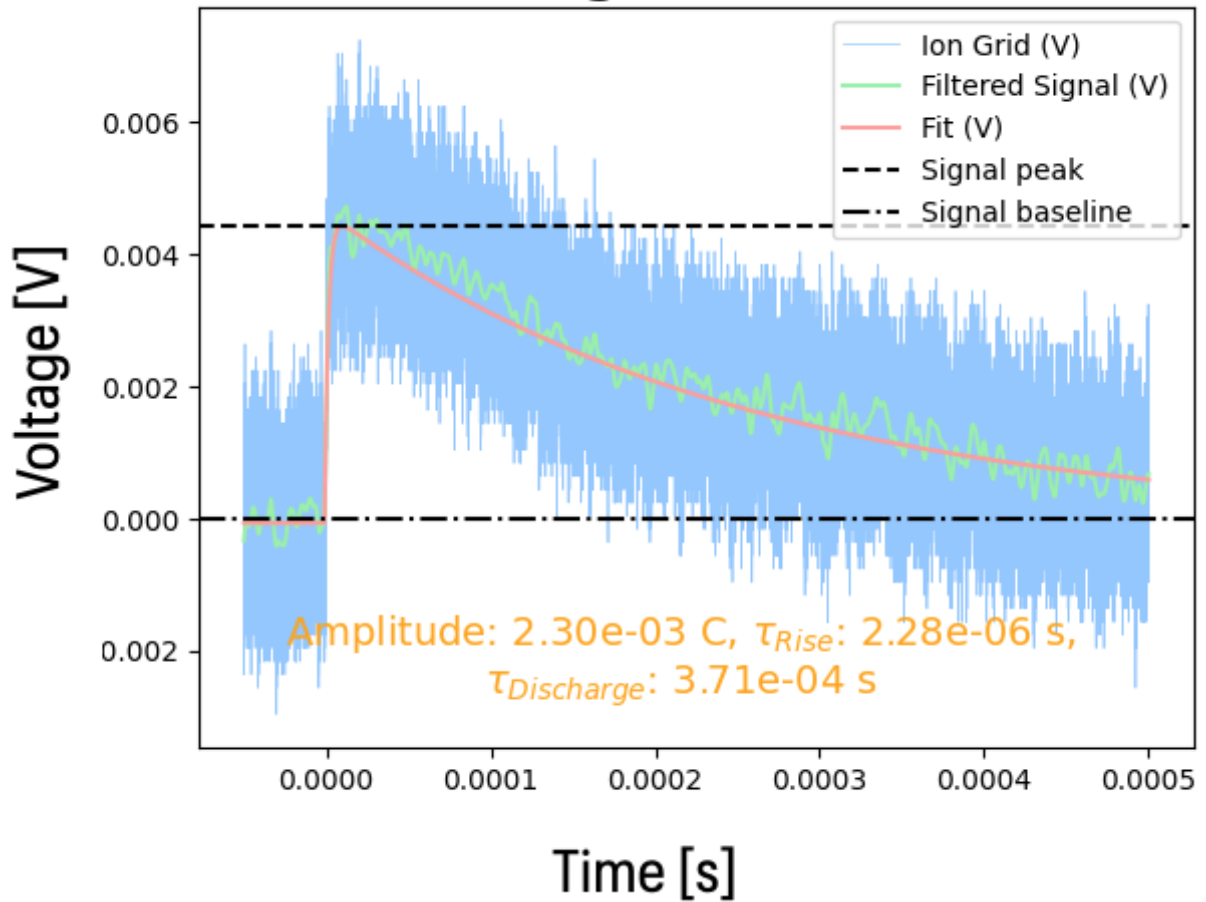


Figure 4.3: An example fit to the target signal. The original signal is blue, the filtered signal is given in green and the fit is the red curve. The fit parameters are displayed in orange text.

Prior to performing the target signal fit, the two gain stages (target low gain, target high gain) have to be combined into one signal. This is done by subtracting an average of the first 10 points from both signals to correct for the offsets in [dN]. Then, the low- and high-gain target signals are scaled by their respective pC/dN conversion values, given in [Table 4.2](#). An example result where the target high gain channel saturates and is shown in [Figure 4.4](#).

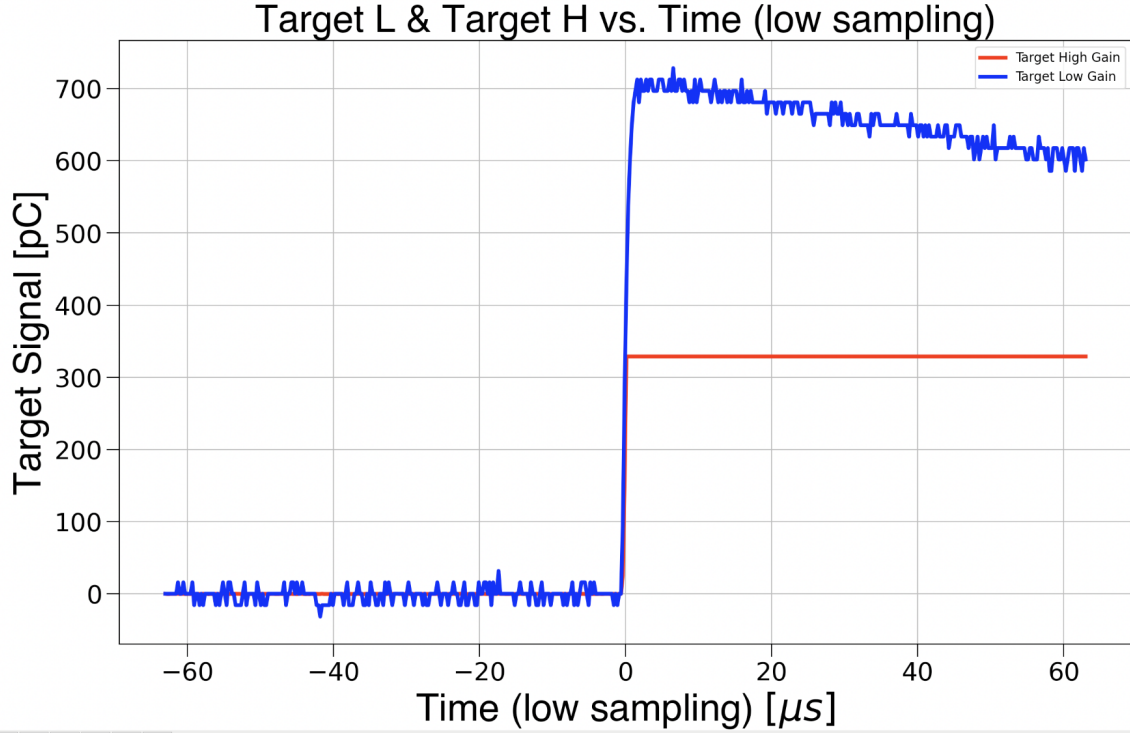


Figure 4.4: An example of combined target signals from Aluminum (Al) dust testing on the IDEX FM with the FM CSAs.

Here, the target low gain (blue) is considered, as the high gain (red) saturates at 328.8 pC. The pC values (**in the specific case of the flight model with the flight CSAs**) were obtained from [Table 4.2](#) and are .163 and 15.8357 [pC/dN] for the high and low gain channels, respectively.

Designate the best waveform (single or combined) with a flag for how it was created ("fit data"). Put this in a flag that can represent any permutation.

The dust mass can be estimated from the numerical integral of the combined TOF spectrum and signal fits to combined target and ion grid channels. The dust particle's impact charge is proportional to both its mass and velocity (Collette, 2014). It is estimated by the amplitude of both the target and ion grid signals, and by the integral of the TOF spectrum. The total charge (Q_i) generated during a dust impact of mass m , at velocity v follows the power law

$$Q_i \propto m^\alpha v^\beta, \quad (4.3)$$

where the exponents α and β vary with the composition of the target and projectile and are experimentally derived. Typical values for α and β fall between 1-2 and 2-5, respectively.

$$Y(v; A, a_1, a_2, a_3, v_b, v_c, k) = A \cdot v^{a_1} \left[\left(1 + \left(\frac{v}{v_b} \right)^k \right)^{\frac{a_2 - a_1}{k}} \cdot \left(1 + \left(\frac{v}{v_c} \right)^k \right)^{\frac{a_3 - a_2}{k}} \right], \quad (4.4)$$

During calibration at the dust accelerator, the impact charge as measured by the amplitude of the target signal was compared to the mass and velocity reported by the accelerator metadata.

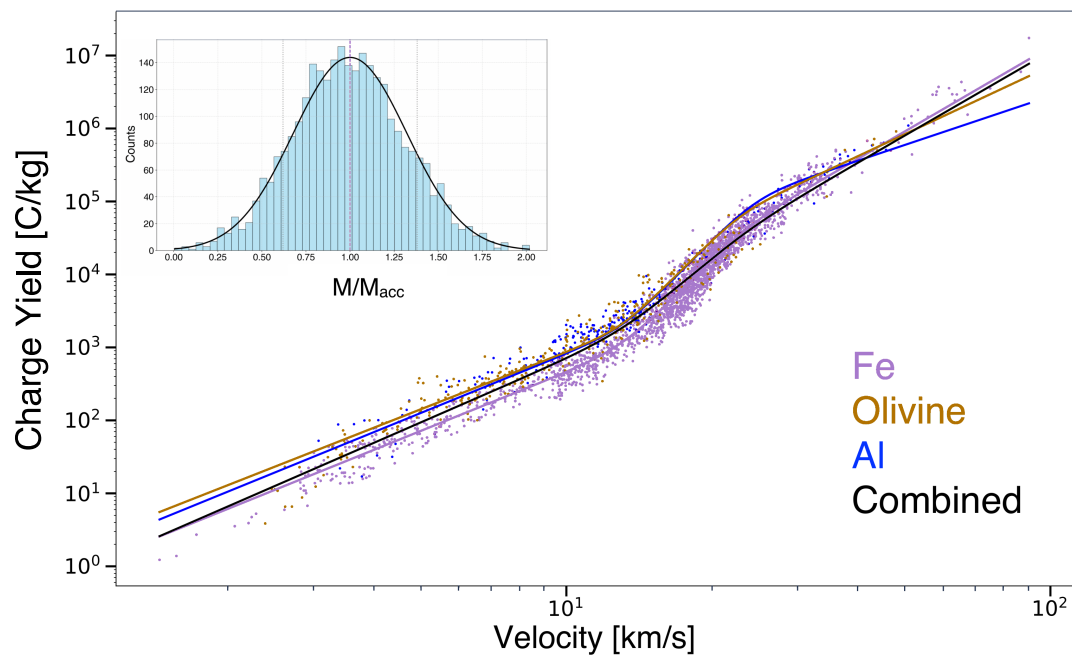


Figure 4.5: Charge yield for iron (red), aluminum (blue), and olivine (green dots) dust particles. The insert shows the distribution of the ratio of the dust particle's mass, as calculated from the fit (Equation 4.4) for all materials combined, over the mass reported by the accelerator.

The fit parameters for each material used to calibrate IDEX are given below.

Material	QT/M								
	A	a_1	a_2	a_3	v_b [km/s]	v_c [km/s]	k	σ	ΔM_Y
Fe	-0.02	2.7	6.7	3.9	13.0	23.7	11.6	0.35	1.41
Olivine	0.32	2.6	6.7	3.1	13.0	22.8	13.9	0.32	1.37
Al	0.21	2.7	6.8	2.2	13.1	24.6	12.3	0.43	1.51
Combined	0.06	2.8	5.9	4.1	13.0	22.7	8.2	0.40	1.47

Material	v								
	A	a_1	a_2	a_3	QT_τ^b [μ s]	QT_τ^c [μ s]	k	σ	Δv_{QT}
Combined	3.6	-0.2	-2.0	0.38	5.1	13.7	13.3	0.28	1.33

Table 4.3: The parameters of the fit of the charge yield as function of the impact speed $Y(v; A, a_1, a_2, a_3, v_b, v_c, k)$ (*top*) and the impact speed as function of the target signal rise time $v(QT_\tau; A, a_1, a_2, a_3, v_b, v_c, k)$ (*bottom*) using the same form Equation 4.4, the standard error σ of the Gaussian fitted to the ratio of the fit over the data, and the error factor of the fit $\Delta M_Y \simeq \exp(\sqrt{\ln(1 + \sigma^2)})$.

In addition to our charge yield curve, our calibration campaign also consisted of developing trends for the target rise time and target to ion grid collection efficiency as a function of speed.

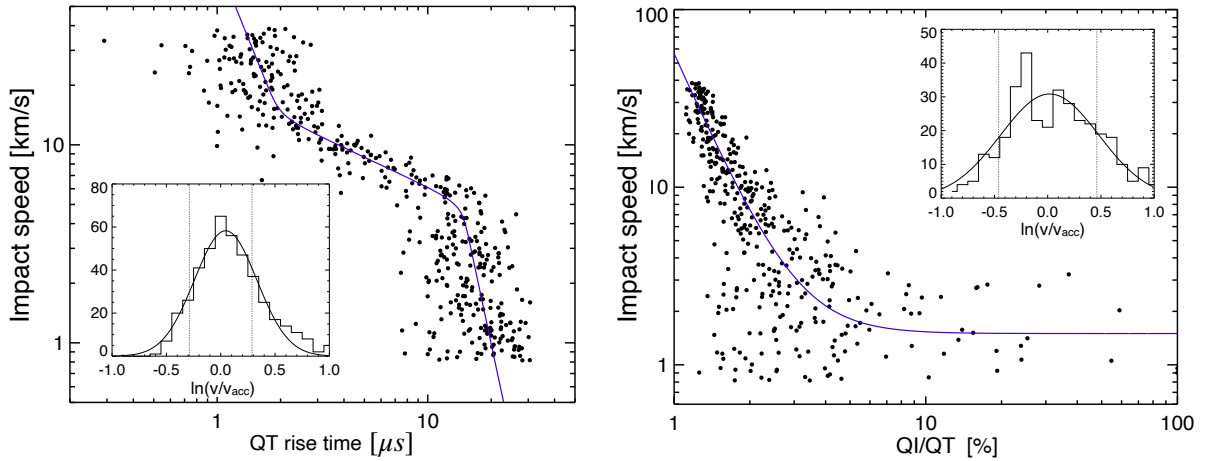


Figure 4.6: The impact speed of a dust particle as a function of the rise time of QT (*left*) and the ratio QI/QT (*right*). The inserts show the Gaussian fits to the log of the ratios of the fit over the data, with $\sigma_{QT} \simeq 0.28$ and $\sigma_{QI/QT} \simeq 0.46$, the error factors (e^σ) are $\Delta v_{QT} \simeq 1.33$ and $\Delta v_{QI/QT} \simeq 1.58$.

This calibration curve is the result of both iron and aluminum dust impacts on the IDEX engineering model (EM) using the ground support equipment (GSE) CSAs. It is used at level 2A to provide an estimate for the dust mass with an error factor ≤ 2 .

In addition to fitting the target and ion grid signals for impact charge, the TOF mass spectrum's x-axis is converted from time to mass. The algorithm to do so is delineated as follows:

1. Make a vector with all zeros and a length of 8189, same as the TOF length of record: t_i .
2. Make a vector for masses m_i : 1, 2, 3, 4, ..., 200 (200 length, and the elements are the integers 1, 2, 3, ..., 200).
3. Calculate the 200 times (T_i -s) these masses should show up for a $T_{\text{off}} = 0$ and stretch A (best guess):

$$T_i = T_{\text{off}} + A\sqrt{m_i}$$

and set the closest $t_i = 1$ to each of the calculated T_i -s, leaving the rest as zero.

4. Calculate the cross-correlation with the original TOF; the maximum will give you the best lag (T_{off}) for a given A .
5. Use an extrema finder to determine the best A .

4.7 Level 2A to 2B Processing

For the level 2B data product, a spectrogram of the number of dust hits as a function of both mass and time will be generated. The simulation data depicted in [Figure 4.2](#) can be used to generate example files for these. The dataset records the count rate of detected dust particle impacts as a function of several key parameters:

- **Time of Impact** (t) - The timestamp of each dust detection event.
- **Spin Phase** (ϕ) - The spacecraft spin phase at the time of detection. This is given in quadrants (0, 90, 180, 270).
- **Mass or Charge** (m/q) - The estimated mass or charge of the detected dust particle.
- **Count rate** ($N/\Delta t_{\text{acc}}$) - The number of detected dust particles in a given bin per accumulation time t_{acc} , where the accumulation time is the width of the time bin subtracted by any dead times.

Science Acquisition Timestamp Extraction

The Science Data Center (SDC) uses IDEX event telemetry to determine periods of science data collection. The IDEX instrument generates event files containing operational state information that are ingested and decommutated daily by the SDC.

The Level 2B (L2B) processing pipeline is triggered monthly and includes all event CDF files from the preceding month plus a two-week buffer period. This span ensures complete coverage for determining instrument operational states across processing boundaries.

The SDC processes event datasets through the following steps:

1. **Data Preparation:** Event data are sorted chronologically by epoch timestamp, and duplicate entries are removed to ensure temporal consistency.
2. **State Change Event Identification:** Science state-change events are identified by filtering for event packets with identifier SCI_STE (Science State Event).
3. **Parameter Reconstruction:** For each state-change event, two 16-bit parameter values are reconstructed by combining pairs of 8-bit telemetry parameters:

- $val1 = (el1par_evtpkt \ll 8) \mid el2par_evtpkt$
- $val2 = (el3par_evtpkt \ll 8) \mid el4par_evtpkt$

4. **Acquisition State Classification:** State transitions are classified based on parameter value combinations:

- *Science Start:* $val1=ACQSETUP$ and $val2=ACQ$ indicate transition to active data acquisition.
- *Science Stop:* $val1=ACQ$ and $val2=CHILL$ indicate transition to idle/standby state.

The charge and mass estimates are placed into 10 bins for the Level 2B IDEX data product. These bins and their ranges are given in table 4.4.

Bin	Mass range [kg]	Charge range [pC]
1	$6.31 \times 10^{-17} - 1.00 \times 10^{-16}$	$1.00 \times 10^{-1} - 3.16 \times 10^{-1}$
2	$1.00 \times 10^{-16} - 1.58 \times 10^{-16}$	$3.16 \times 10^{-1} - 1.00 \times 10^0$
3	$1.58 \times 10^{-16} - 2.51 \times 10^{-16}$	$1.00 \times 10^0 - 3.16 \times 10^0$
4	$2.51 \times 10^{-16} - 3.98 \times 10^{-16}$	$3.16 \times 10^0 - 1.00 \times 10^1$
5	$3.98 \times 10^{-16} - 6.31 \times 10^{-16}$	$1.00 \times 10^1 - 3.16 \times 10^1$
6	$6.31 \times 10^{-16} - 1.00 \times 10^{-15}$	$3.16 \times 10^1 - 1.00 \times 10^2$
7	$1.00 \times 10^{-15} - 1.58 \times 10^{-15}$	$1.00 \times 10^2 - 3.16 \times 10^2$
8	$1.58 \times 10^{-15} - 2.51 \times 10^{-15}$	$3.16 \times 10^2 - 1.00 \times 10^3$
9	$2.51 \times 10^{-15} - 3.98 \times 10^{-15}$	$1.00 \times 10^3 - 3.16 \times 10^3$
10	$3.98 \times 10^{-15} - 1.00 \times 10^{-14}$	$3.16 \times 10^3 - 1.00 \times 10^4$

Table 4.4: Bin definitions for mass and charge used for IDEX L2B data products.

For the edge case that a mass or charge value is below the value

The following table presents a sample of the Level 2B data product. The values represent binned counts of detected dust particles:

Please note that:

- Times are given in Coordinated Universal Time (UTC).
- Spin phase is defined relative to an inertial (despun) reference frame.
- Mass and charge values are estimated using calibration data.

Table 4.5: Daily Binned Dust Hit Counts as a Function of Time, Spin Phase, and Mass/Charge

Date (UTC)	Spin Phase (deg)	Mass/Charge (kg or C)	Rate (s^{-1})
2027-02-18	0	1.2×10^{-15}	32
2027-02-19	90	3.5×10^{-16}	60
2027-02-20	180	2.51×10^{-15}	25
2027-02-21	270	4.8×10^{-16}	48

4.8 Level 2A to 2C Processing

Generate pointing sets from level 2A. All events will be dust, so pointing sets follow directly. Count rates are encoded in the pointing sets. We want RTN for IDPs, EJ2000 for ISD. A time-ordered list of time, boresight vector for every impact. We will also need accumulation times. Disregard events that are flagged as SW triggers.

For the level 2C data products, maps of flux, rates and composition will be generated, using the IBEX ENA maps as reference. An example of a historical dust map for the interstellar dust is shown in [Figure 4.7](#).

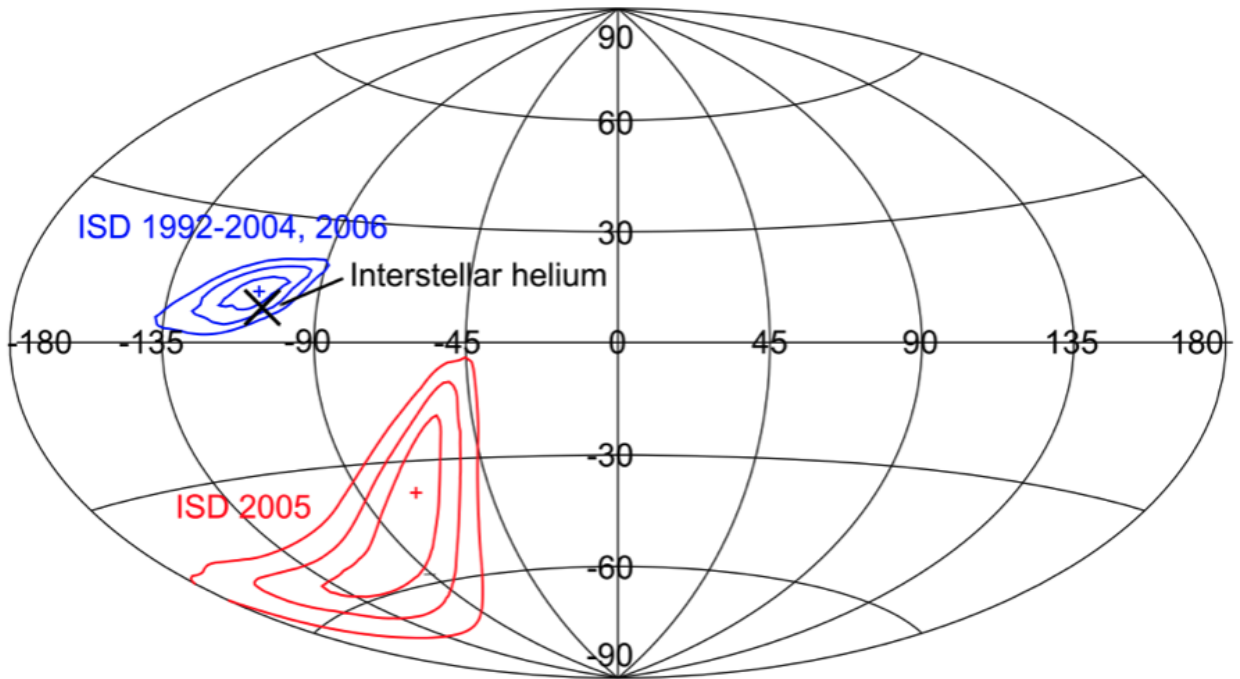


Figure 4.7: Map of the two different deflections of interstellar dust measured by Cassini's CDA and Ulysses from the nominal interstellar helium flow direction into the solar system.

These maps will require defining **pointing sets**, which contain the data binned by instrument boresight vector. For IDEX maps, each dust impact is assumed to come directly from the boresight direction (neglecting the 50-degree field-of-view).

Typical data for a 2 year mission are shown in figure 4.8. The y and x axes represent the EJ2000 latitude and longitude, respectively. The incoming ISD direction is apparent from the enhancement at (8, -118).

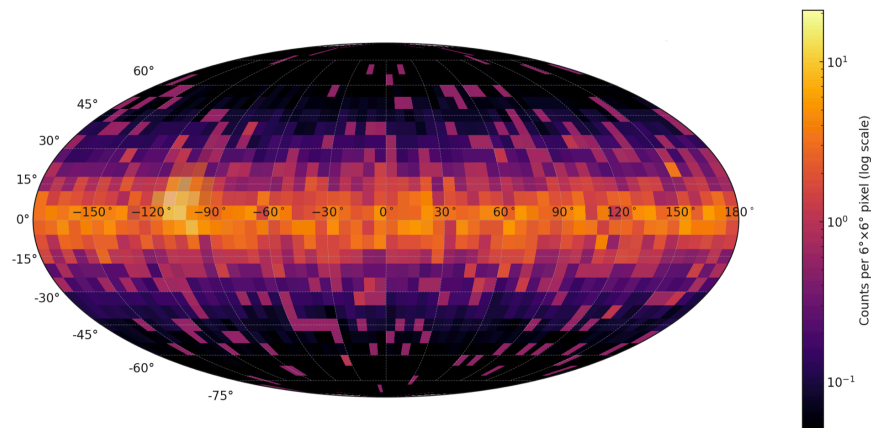


Figure 4.8: Total dust counts per 6 x 6 degree pixels in latitude and longitude in EJ2000. This is assuming a dust instrument mounted to a spacecraft at 1 AU, such as IDEX. These maps will be used for direct comparison of the COMIT model's results with IMAP results. In addition to total dust counts, they will be generated for a given impact charge, mass and dust composition type (silicate, organic, etc.) for further comparison with IDEX.

Appendix A

LASP Packets

The **generator** method from the **parser.py** script in LASP Packets.

```
1 def generator(self, binary_data: bitstring.ConstBitStream, parse_bad_pkts: bool = True,
  skip_header_bits: int = 0, root_container_name="CCSDSPacket", ccsds_headers_only: bool =
  False, yield_unrecognized_packet_errors: bool = False, show_progress: bool = False):
2     """Create and return a Packet generator.
3     Creating a generator object to return allows the user to create many generators from a
  single Parser and reduces memory usage.
4     ...
```

Appendix B

IDEX/SUDA Combined XTCE Definition

The first 100 lines of the combined IDEX/SUDA XTCE definition are shown below.

```
1 <?xml version="1.0" encoding="UTF-8"?>
2 <xtce:SpaceSystem xmlns:xtce="http://www.omg.org/space/xtce" name="IDEX">
3   <xtce:Header classification="BuildType=FLIGHT, CreatedBy=XTCEEditor" date="Created
   =2023/01/12 15:27:45" version="DocumentVersion=FSW Build ,DocumentVersion=20190109
   v1,LASPXTCESchemaVersion=1.2,XTCEEditorVersion=0.47"/>
4   <xtce:TelemetryMetaData>
5     <xtce:ParameterTypeSet>
6       <xtce:IntegerParameterType signed="false" name="VERSION_Type">
7         <xtce:UnitSet/>
8         <xtce:IntegerDataEncoding encoding="unsigned" sizeInBits="3"/>
9       </xtce:IntegerParameterType>
10      <xtce:IntegerParameterType signed="false" name="TYPE_Type">
11        <xtce:UnitSet/>
12        <xtce:IntegerDataEncoding encoding="unsigned" sizeInBits="1"/>
13      </xtce:IntegerParameterType>
14      <xtce:IntegerParameterType signed="false" name="SEC_HDR_FLG_Type">
15        <xtce:UnitSet/>
16        <xtce:IntegerDataEncoding encoding="unsigned" sizeInBits="1"/>
17      </xtce:IntegerParameterType>
18      <xtce:IntegerParameterType signed="false" name="PKT_APID_Type">
19        <xtce:UnitSet/>
20        <xtce:IntegerDataEncoding encoding="unsigned" sizeInBits="11"/>
21      </xtce:IntegerParameterType>
22      <xtce:IntegerParameterType signed="false" name="SEQ_FLGS_Type">
23        <xtce:UnitSet/>
24        <xtce:IntegerDataEncoding encoding="unsigned" sizeInBits="2"/>
25      </xtce:IntegerParameterType>
26      <xtce:IntegerParameterType signed="false" name="SRC_SEQ_CTR_Type">
27        <xtce:UnitSet/>
```

```

28     <xtce:IntegerDataEncoding encoding="unsigned" sizeInBits="14"/>
29 </xtce:IntegerParameterType>
30 <xtce:IntegerParameterType signed="false" name="PKT_LEN_Type">
31     <xtce:UnitSet/>
32     <xtce:IntegerDataEncoding encoding="unsigned" sizeInBits="16"/>
33 </xtce:IntegerParameterType>
34 <xtce:FloatParameterType name="SHCOARSE_Type">
35     <xtce:UnitSet>
36         <xtce:Unit>dn</xtce:Unit>
37     </xtce:UnitSet>
38     <xtce:IntegerDataEncoding encoding="unsigned" sizeInBits="32"/>
39 </xtce:FloatParameterType>
40 <xtce:FloatParameterType name="SHFINE_Type">
41     <xtce:UnitSet>
42         <xtce:Unit>dn</xtce:Unit>
43     </xtce:UnitSet>
44     <xtce:IntegerDataEncoding encoding="unsigned" sizeInBits="16"/>
45 </xtce:FloatParameterType>
46 <xtce:FloatParameterType name="CHECKSUM_Type">
47     <xtce:UnitSet>
48         <xtce:Unit>dn</xtce:Unit>
49     </xtce:UnitSet>
50     <xtce:IntegerDataEncoding encoding="unsigned" sizeInBits="32"/>
51 </xtce:FloatParameterType>
52 <xtce:IntegerParameterType name="IDX__SCI0AID_Type">
53     <xtce:UnitSet/>
54     <xtce:IntegerDataEncoding encoding="unsigned" sizeInBits="32"/>
55 </xtce:IntegerParameterType>
56 <xtce:IntegerParameterType name="IDX__SCI0TYPE_Type">
57     <xtce:UnitSet/>
58     <xtce:IntegerDataEncoding encoding="unsigned" sizeInBits="8"/>
59 </xtce:IntegerParameterType>
60 <xtce:IntegerParameterType name="IDX__SCI0CONT_Type">
61     <xtce:UnitSet/>
62     <xtce:IntegerDataEncoding encoding="unsigned" sizeInBits="8"/>
63 </xtce:IntegerParameterType>
64 <xtce:IntegerParameterType name="IDX__SCI0SPARE1_Type">
65     <xtce:UnitSet/>
66     <xtce:IntegerDataEncoding encoding="unsigned" sizeInBits="13"/>
67 </xtce:IntegerParameterType>
68 <xtce:EnumeratedParameterType name="IDX__SCI0PACK_Type">
69     <xtce:UnitSet/>
70     <xtce:IntegerDataEncoding encoding="unsigned" sizeInBits="1"/>
71     <xtce:EnumerationList>
72         <xtce:Enumeration value="0" label="DS"/>
73         <xtce:Enumeration value="1" label="EN"/>
74     </xtce:EnumerationList>
75 </xtce:EnumeratedParameterType>

```

```

76 <xtce:EnumeratedParameterType name="IDX__SCI0FRAG_Type">
77   <xtce:UnitSet/>
78   <xtce:IntegerDataEncoding encoding="unsigned" sizeInBits="1"/>
79   <xtce:EnumerationList>
80     <xtce:Enumeration value="0" label="DS"/>
81     <xtce:Enumeration value="1" label="EN"/>
82   </xtce:EnumerationList>
83 </xtce:EnumeratedParameterType>
84 <xtce:EnumeratedParameterType name="IDX__SCI0COMP_Type">
85   <xtce:UnitSet/>
86   <xtce:IntegerDataEncoding encoding="unsigned" sizeInBits="1"/>
87   <xtce:EnumerationList>
88     <xtce:Enumeration value="0" label="DS"/>
89     <xtce:Enumeration value="1" label="EN"/>
90   </xtce:EnumerationList>
91 </xtce:EnumeratedParameterType>
92 <xtce:IntegerParameterType name="IDX__SCI0EVTNUM_Type">
93   <xtce:UnitSet/>
94   <xtce:IntegerDataEncoding encoding="unsigned" sizeInBits="16"/>
95 </xtce:IntegerParameterType>
96 <xtce:IntegerParameterType name="IDX__SCI0CAT_Type">
97   <xtce:UnitSet/>
98   <xtce:IntegerDataEncoding encoding="unsigned" sizeInBits="8"/>

```